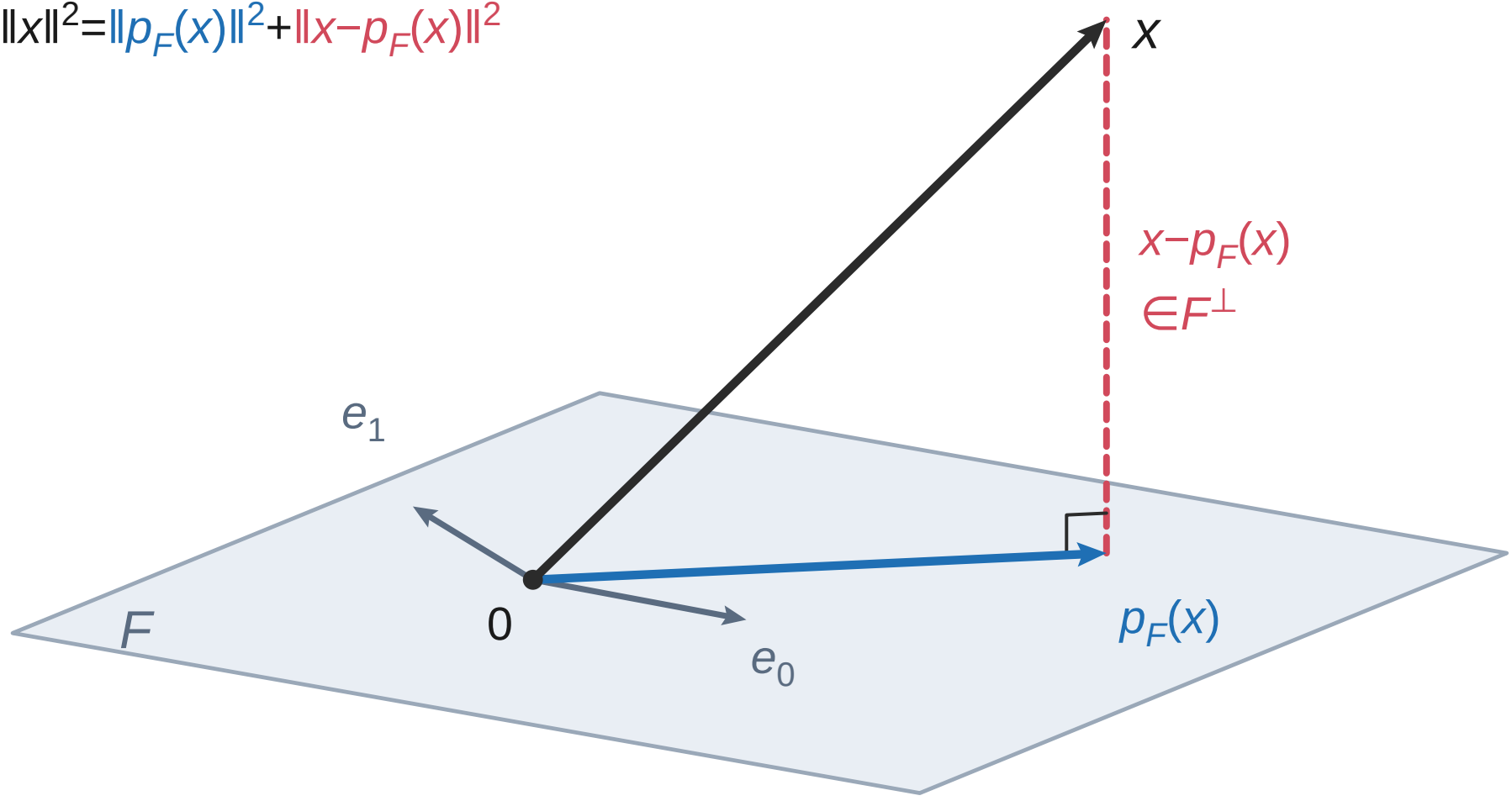$\|x\|^2 = \|p_F(x)\|^2 + \|x - p_F(x)\|^2$